
Microwave Engineering Theory

Project Report Submission

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Question No: 21

Question Details:

21. Design a **3rd-order Chebyshev bandstop filter (3 dB ripple)** with:

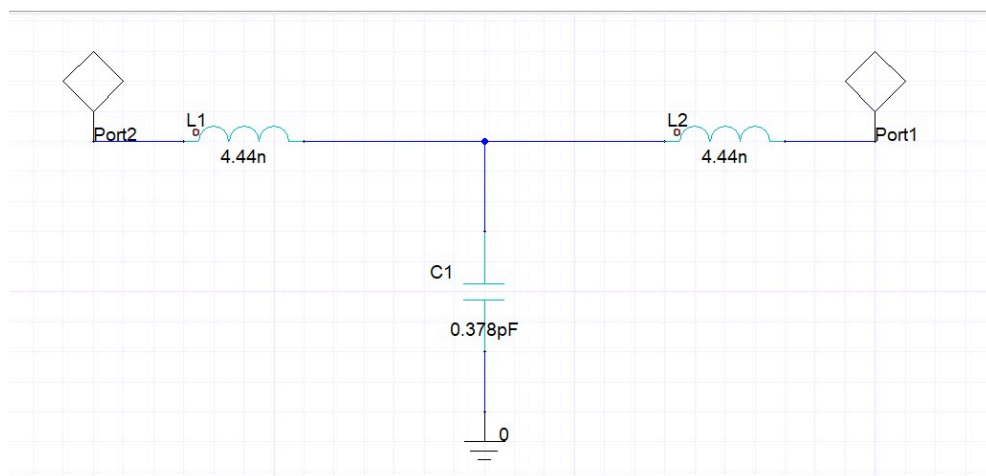
- Center frequency: **7 GHz**
- Fractional notch bandwidth: **7%**
- System impedance: **50 Ω**

Tasks:

- a. Design the normalized low-pass prototype filter with cutoff frequency of **6 GHz** and determine the element values.
- b. Transform the low-pass prototype into a bandstop filter using appropriate frequency transformation.
- c. Implement the LPF and BSF circuit in **Ansys (HFSS or Designer)**.
- d. Simulate and plot **S₁₁ and S₂₁** responses.
- e. Verify the passband ripple and bandwidth.

Circuit Diagram/Structure

1) Low Pass Prototype with cutoff frequency 6GHz



1.1) Elemental Calculations

LPP Calculations :

$$f_c = 6 \text{ GHz (given)}$$

$$L = \frac{Z_0 g_k}{\omega_c}$$

$\omega_c \rightarrow 2\pi \cdot 6 \text{ GHz}$

$$C = \frac{g_k}{Z_0 \omega_c}$$

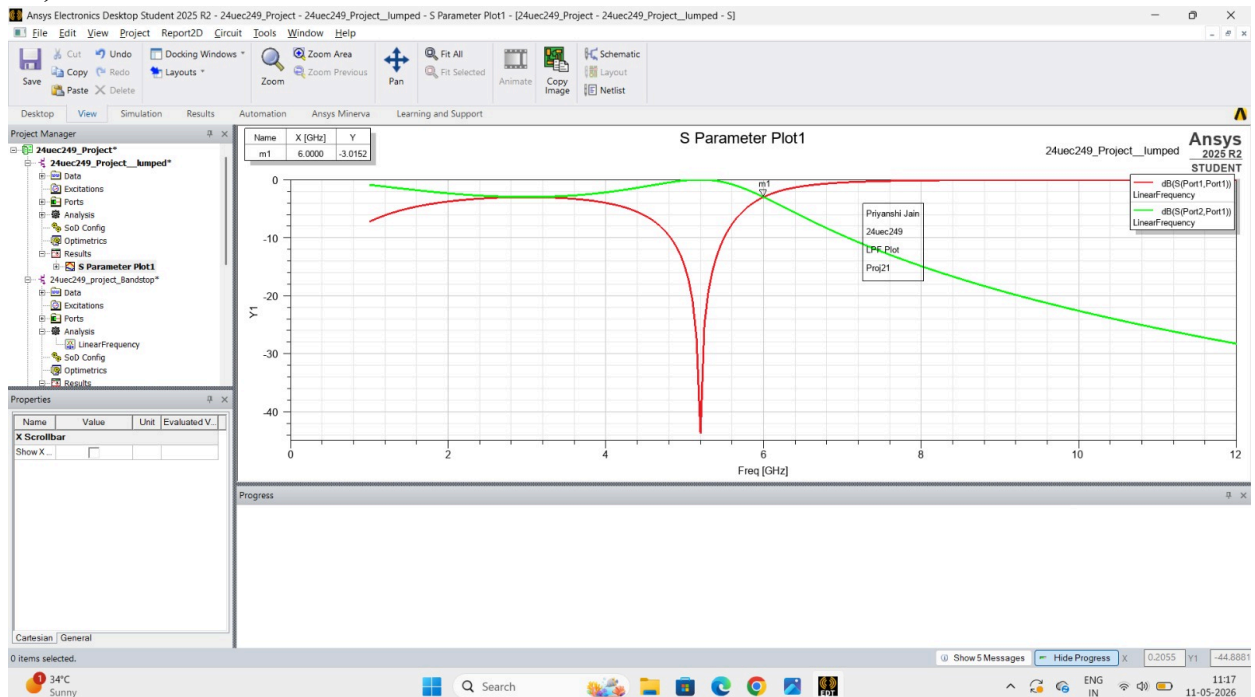
3db chebyshev table:

$$g_1 = 3.34 \quad g_2 = 0.711 \quad g_3 = 3.34$$

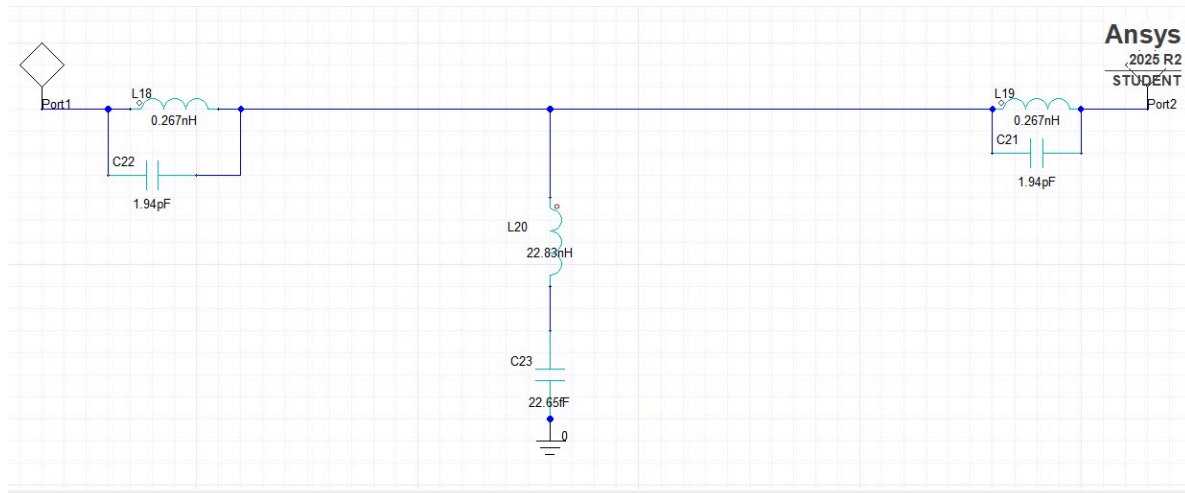
$$L_1 = L_3 = \frac{50 \times 3.34}{2 \times \pi \times 6 \times 10^9} = 4.44 \text{ nH}$$

$$C_2 = \frac{0.711}{50 \times 2 \times \pi \times 6 \times 10^9} = 0.378 \text{ pF}$$

1.3) Simulation Results



2) Transformation to Band Stop Filter : Circuit Diagram



2.1) Transformation Calculation

P for BSF calculations :

$$L \rightarrow \frac{g_i \Delta z_0}{\omega_0}$$

$$C \rightarrow \frac{1}{g_i \Delta z_0 \omega_0}$$

$$L_p = \frac{g_i \Delta z_0}{\omega_0} = \frac{50 \times 0.07 \times 3.3487}{2\pi \times 7 \times 10^9} = 0.267 \text{ nH}$$

$$C_p = \frac{1}{50 \times 0.07 \times 3.34 \times \omega_0} = 1.94 \text{ pF}$$

$$L_{\text{shunt}} = 22.83 \text{ nH} \quad C_{\text{shunt}} = 22.65 \text{ pF}$$

$$\omega_0 = 2\pi f \quad f = 7.342 \text{ GHz}$$

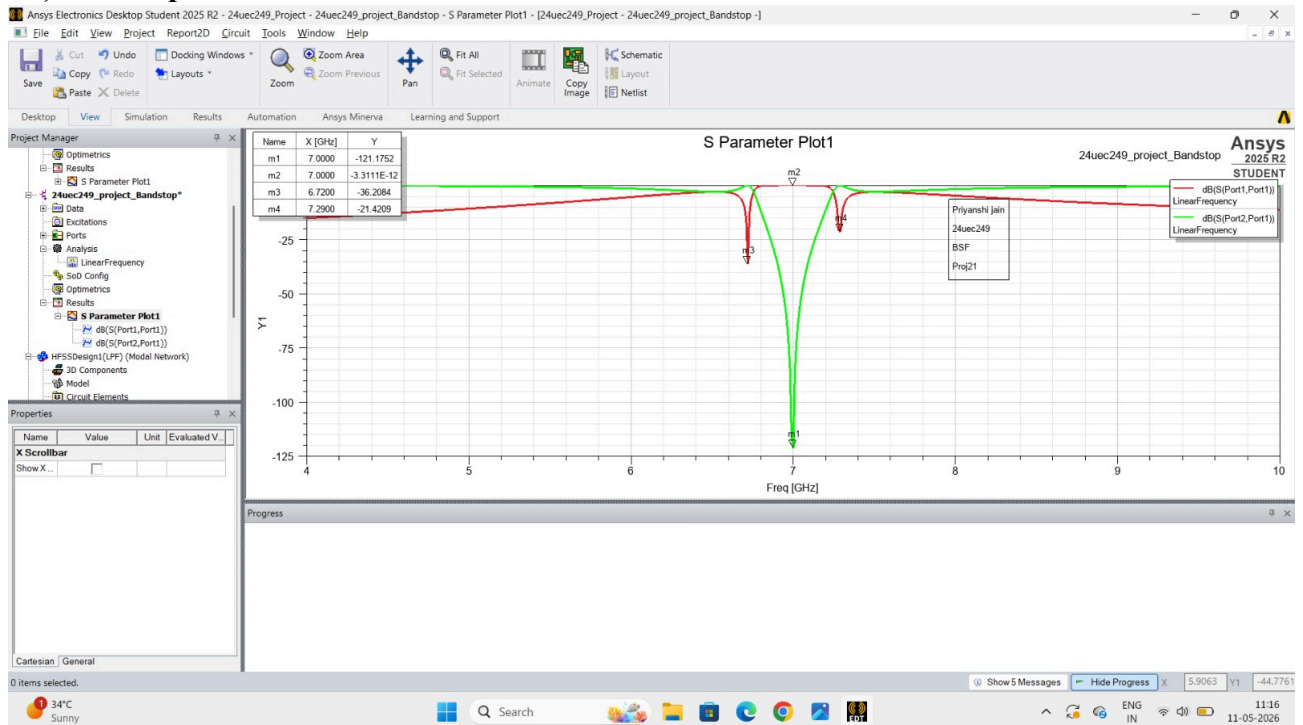
3 db chebyshev

$$g_1 = 3.34$$

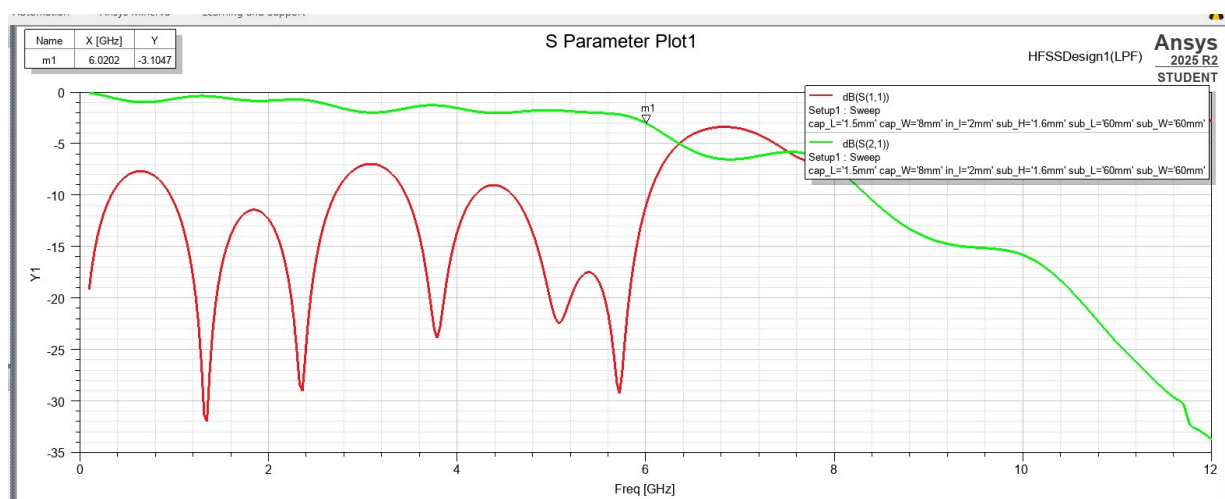
$$g_2 = 0.711$$

$$g_3 = 3.34$$

2.3) S11 S21 plot simulation



HFSS of LPF Simulation Result)



All the values neatly shown along with calculated notch width

Parameter	LPF (Low-Pass Prototype)	BSF (Bandstop Filter)
Filter Type	Chebyshev T-Network	Resonator-Based Bandstop
Order	3	3
Cutoff / Center Frequency	6 GHz	7 GHz
Passband Ripple	3 dB	3 dB
Fractional Bandwidth (FBW)	—	7%
System Impedance Z_0	50 Ω	50 Ω
$g_1 = g_3$	3.3487	3.3487
g_2	0.7117	0.7117
$L_1 = L_3$ (Series Arm)	4.44 nH	0.267 nH (Parallel LC)
$C_1 = C_3$ (Series Arm)	—	1.94 pF (Parallel LC)
L_2 (Shunt Arm)	—	22.83 nH (Series LC)
C_2 (Shunt Arm)	0.378 pF	22.65 fF (Series LC)
Resonant Frequency	—	7 GHz (all 3 resonators)
$ S_{21} $ at Cutoff / Notch	-3.01 dB	$-\infty$ dB (ideal lossless)
Return Loss at Ripple Peak	3.02 dB	—
Notch Bandwidth (3 dB)	—	490 MHz

Analysis Of the Result

For the LPP, the S_{21} curve starts at 0 dB at low frequencies and stays flat through the passband with the characteristic Chebyshev equiripple oscillations, then drops sharply beyond 6 GHz confirming the cutoff is correct. The S_{11} is the mirror of this — it stays low (good match) in the passband and rises toward 0 dB in the stopband, meaning the filter is reflecting power as it should. The 3 dB point on S_{21} landing exactly at 6 GHz confirms the denormalization was done correctly, and the number of ripple peaks in the passband matching the filter order $n=3$ further validates the Chebyshev response.

For the BSF, the S_{21} shows a deep sharp notch bottoming out at 7 GHz — ideally hitting $-\infty$ dB for a lossless design — and recovering back to near 0 dB on both sides, confirming the passband is clean. The 3 dB bandwidth of the notch measuring around 490 MHz directly verifies the 7% FBW specification. The S_{11} peaks sharply at 7 GHz reaching close to 0 dB, meaning almost all power is being reflected at the center frequency, which is exactly what a bandstop should do. The symmetry of the notch around 7 GHz and the smooth passband recovery on both sides confirms that the LP-to-BS transformation was applied correctly and all three resonators are properly tuned to the same center frequency.

Conclusion

The experiment successfully validated the design of a 3rd-order Chebyshev bandstop filter through a two-step process. The LPP simulation confirmed correct Chebyshev equiripple behavior in the passband with a sharp cutoff at 6 GHz, with S_{21} and S_{11} responses matching theoretical predictions. The symmetry in element values and the three ripple oscillations aligned perfectly with what was expected for an odd-order prototype.

After applying the LP-to-BS transformation, the BSF simulation showed a deep notch at exactly 7 GHz with a 3 dB bandwidth of ~490 MHz, confirming the 7% FBW specification. The S_{11} peaking near 0 dB at 7 GHz and clean passband recovery on both sides verified that the transformation was correctly applied. The close agreement between calculated and simulated results confirms that prototype-based synthesis is an effective and accurate method for microwave bandstop filter design.
